

Multiple Choice

1. **Answer:** C

Options: A) $101.1 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; B) $26.75 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; C) $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; D) $0.9377 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$

Note: Correct option: C - $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$

2. **Answer:** D

Options: A) B; B) Al; C) In; D) Tl

Note: Correct option: D - Tl

3. **Answer:** A

Options: A) Neutrons; B) Alpha particles; C) Beta particles; D) X-rays

Note: Correct option: A - Neutrons

4. **Answer:** D

Options: A) The process is exothermic.; B) The process does not involve any work.; C) The entropy of the system increases.; D) The total entropy of the system plus surroundings increases.

Note: Correct option: D - The total entropy of the system plus surroundings increases.

5. **Answer:** D

Options: A) $6.345 \times 10^{-4} \text{ at } 0.335 \text{ T}$; $9.871 \times 10^{-5} \text{ at } 10.5 \text{ T}$; B) $0.793 \times 10^{-4} \text{ at } 0.335 \text{ T}$; $6.931 \times 10^{-7} \text{ at } 10.5 \text{ T}$; C) $4.126 \times 10^{-3} \text{ at } 0.335 \text{ T}$; $2.142 \times 10^{-6} \text{ at } 10.5 \text{ T}$

Note: Correct option : D - $4.126 \times 10^{-3} \text{ at } 0.335 \text{ T}$; $2.142 \times 10^{-6} \text{ at } 10.5 \text{ T}$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Potassium hydrogen phthalate'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It is a red-brown volatile liquid.'

Long Form Response

11. **Answer:** hexane. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 2-methylpentane. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key